

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FIFTH SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2018

Course Code: EE309

Course Name: MICROPROCESSOR AND EMBEDDED SYSTEMS

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 5 marks.

Marks

- | | | |
|---|---|-----|
| 1 | Explain the following pins in 8085 Microprocessor.
i) HOLD ii) READY iii) TRAP iv) ALE | (5) |
| 2 | Explain how delay is generated using one register and calculate how much delay can be generated if one register is loaded with maximum count. Assume time for one state is 320ns. | (5) |
| 3 | Explain the interrupts in 8085 Microprocessor. | (5) |
| 4 | Give comparison between Microprocessors and Microcontrollers. | (5) |
| 5 | Explain the following instructions used in 8051 microcontroller.
i) MOV R ₁ , #05 _H ii) ADD A, #01 _H iii) MOV R ₂ , 07 _H iv) DJNZ R ₂ , loop | (5) |
| 6 | Explain with neat diagram the RAM of 8051. | (5) |
| 7 | Explain the Data types and Directives of 8051 Microcontroller. | (5) |
| 8 | Write a program to create a square wave of 50% duty cycle on the P1.5 bit .
Timer 0 is used to generate the time delay. | (5) |

PART B

Answer any two full questions, each carries 10 marks.

- | | | |
|----|--|------|
| 9 | a) Write an ALP in 8085 to find the smallest number in an array | (5) |
| | b) Draw the timing diagram for Memory Read operation. | (5) |
| 10 | Explain with neat block diagram the architecture of Intel 8085 Microprocessor. | (10) |
| 11 | a) Explain CALL & RETURN instructions in 8085. | (5) |
| | b) Explain Machine Cycle and Instruction Cycle | (5) |

PART C

Answer any two full questions, each carries 10 marks.

- | | | |
|----|--|-----|
| 12 | a) Explain with suitable diagram, how an ADC can be interfaced with 8085 Microprocessor. | (6) |
| | b) Explain Software and Hardware interrupts in 8085 Microprocessor. | (4) |

- 13 a) Draw an Interface circuit of 2048 x 8 bit RAM with 8085 using NAND Gate. (5)
b) Explain general characteristics of Embedded system. (5)
- 14 a) Explain i) Compiler ii) Assembler iii) Linker iv) Loaders. (8)
b) What are the demerits of Waterfall Model? (2)

PART D

Answer any two full questions, each carries 10 marks.

- 15 Explain with neat block diagram the architecture of 8051 Microcontroller. (10)
- 16 a) Explain SFR registers in 8051 microcontroller. (5)
b) Explain TMOD and TCON registers of 8051 Microcontroller. (5)
- 17 Write an ALP in 8051 to add two 32 bit numbers & store the result. (10)

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Scheme of Valuation/Answer Key

(Scheme of evaluation (marks in brackets) and answers of problems/key)

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V SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2018

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Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 5 marks.

		Marks
1	Each (1.25)	(5)
2	Program(3) + delay time= 1.1ms (2)	(5)
3	5 interrupts explanation (5)	(5)
4	Five comparison (5)	(5)
5	Each (1.25)	(5)
6	Diagram (3)+ Explanation (2)	(5)
7	Data types(2) + Directives (3)	(5)
8	Program (5)	(5)

Students making a positive attempt to answer this question (correct logic/flowchart/steps/algorithm) may be provided with proportional weightage of marks.

PART B

Answer any two full questions, each carries 10 marks.

- 9 a) Program (5) (5)
- Students making a positive attempt to answer this question (correct logic/flowchart/steps/algorithm) may be provided with proportional weightage of marks.*
- b) timing diagram (5) (5)

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- 10 Diagram (4) + Explanation (6) (10)
- 11 a) Explanation (5) (5)
- b) Machine Cycle (2.5) + Instruction Cycle(2.5) (5)

PART C

Answer any two full questions, each carries 10 marks.

- 12 a) Diagram (4) + Explanation (2) (6)
- b) Software interrupts (2) + Hardware interrupts (2) (4)
- 13 a) Diagram (5) (5)
- b) Any five characteristics (5) (5)
- 14 a) Each one 2 marks (2marks x 4=8 marks) (8)
- b) two demerits (2) (2)

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PART D

Answer any two full questions, each carries 10 marks.

- 15 Diagram (5) + Explanation (5) (10)
- 16 a) SFR registers(5) (5)
- b) TMOD registers (2.5) + TCON registers (2.5) (5)
- 17 Program (10) (10)

Students making a positive attempt to answer this question (correct logic/flowchart/steps/algorithm) may be provided with proportional weightage of marks.

QP SOLUTION

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PART A

Answer all questions, each carries 5 marks

1. Explain the following pins in 8085 Microprocessor.

i) HOLD ii) READY iii) TRAP iv) ALE

i) HOLD: Indicates that another Master is requesting the use of the Address and Data Buses. The CPU upon receiving the Hold request will relinquish the use of buses as soon as the completion of the current machine cycle. Internal processing can continue. The processor can regain the buses only after the Hold is removed. When the Hold is acknowledged, the Address, Data, RD, WR, and IO/M lines are 3stated

ii) READY: Ready is an input pin, is high during a read or write cycle, it indicates that the memory or peripheral is ready to send or receive data. If Ready is low, the CPU will wait for Ready to go high before completing the read or write cycle.

iii) TRAP: Trap interrupt is a non-maskable restart interrupt. It is recognized at the same time as INTR. It is unaffected by any mask or Interrupt Enable. It has the highest priority of any interrupt

iv) ALE (Address Latch Enable): It occurs during the first clock cycle of a machine state and enables the address to get latched into the on chip latch of peripherals. The falling edge of ALE is set to guarantee setup and hold times for the address information. ALE can also be used to strobe the status information. ALE is never tristated

2. Explain how delay is generated using one register and calculate how much delay can be generated if one register is loaded with maximum count. Assume time for one state is 320ns.

Delay Program using one register

```
                MVI A, FFH          7T    ; maximum delay is 255d = FFH
Repeat:         DCR A              4T
                JNZ repeat         7T/10T
                RET
```

The total delay = $7T + 255(4T+10T) - 3T$
= 3574T

If one T = 320ns;

Delay = $3574 \times 320\text{ns} = 1143680\text{ns} = 1.1437 \text{ msec}$

3. Explain the interrupts in 8085 Microprocessor.

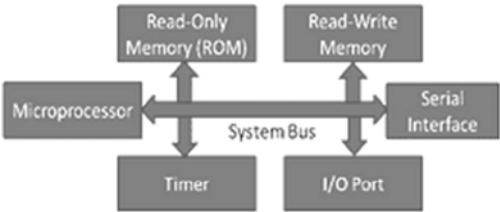
To answer a request from an I/O device, it is sometimes necessary to interrupt the execution of the main program. The CPU temporarily stops the ongoing process, attends the I/O device which has interrupted the CPU, then returns to what it was doing. Interrupt process of 8085 is controlled by the Interrupt Enable Flip Flop, which is internal to the microprocessor. Interrupt Enable Flip Flop can be set or reset by using software instructions

8051 interrupts are INTR, RST 7.5, RST 6.5, RST 5.5, TRAP

Interrupt Classification

- According upon their Maskability
 - i. Maskable Interrupts → INTR, RST 7.5, RST 6.5, RST 5.5
 - ii. Non- maskable interrupts → TRAP
- According their address distribution method.
 - i. Vectored Interrupts → Microprocessor automatically goes to specific address in response to an interrupt request without any external hardware
eg:- TRAP, RST 7.5, RST 6.5, RST 5.5
 - ii. Non-vectored Interrupts → Interrupt device has to give the address of the interrupt service routine
eg:- INTR

4. Give comparison between Microprocessors and Microcontrollers

Microprocessor	Micro Controller
	
Microprocessor is heart of Computer system.	Micro Controller is a heart of embedded system.
It is just a processor. Memory and I/O components have to be connected externally	Micro controller has external processor along with internal memory and i/O components
Since memory and I/O has to be connected externally, the circuit becomes large.	Since memory and I/O are present internally, the circuit is small.
Cannot be used in compact systems and hence inefficient	Can be used in compact systems and hence it is an efficient technique
Cost of the entire system increases	Cost of the entire system is low
Due to external components, the entire power consumption is high. Hence it is not suitable to used with devices running on stored power like batteries.	Since external components are low, total power consumption is less and can be used with devices running on stored power like batteries.
Most of the microprocessors do not have power saving features.	Most of the micro controllers have power saving modes like idle mode and power saving mode. This helps to reduce power consumption even further.
Since memory and I/O components are all external, each instruction will need external operation, hence it is relatively slower.	Since components are internal, most of the operations are internal instruction, hence speed is fast.
Microprocessor have less number of registers, hence more operations are memory based.	Micro controller have more number of registers, hence the programs are easier to write.
Microprocessors are based on von Neumann model/architecture where program and data are stored in same memory module	Micro controllers are based on Harvard architecture where program memory and Data memory are separate
Mainly used in personal computers	Used mainly in washing machine, MP3 players

5. Explain the following instructions used in 8051 microcontroller

i) MOV R1, #05H ii) ADD A, #01H iii) MOV R2, 07H iv) DJNZ R2, loop

- i) MOV R1, #05H : Immediate addressing mode, Load the R1 register immediately with a hexadecimal data 05H, No flags affected
- ii) ADD A, #01H: Immediate addressing mode, Add the content of accumulator with a hexadecimal data 01H and store the result in accumulator, All flags affected
- iii) MOV R2, 07H : Direct addressing mode, Load the content of 07H to R2 register directly, No flags affected
- iv) DJNZ R2, loop: Decrement the R2 register by one and jump to the given location named as “loop” if the content of register R2 is Non-zero.

6. Explain with neat diagram the RAM of 8051

8051 Micro controller contains 128Byte of internal RAM and is divided into three sections.

i. 32 bytes-Register Bank:

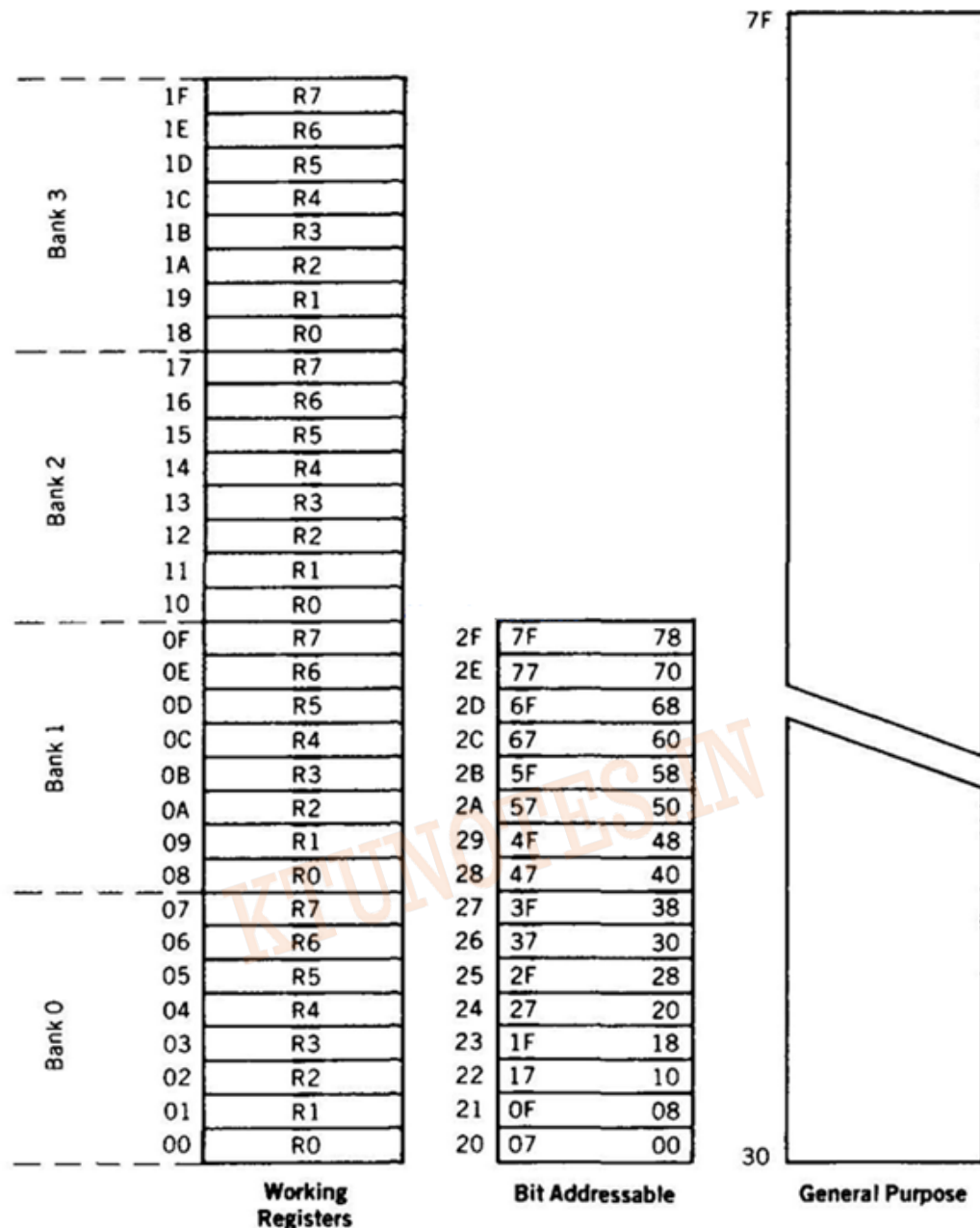
There are four register banks in RAM, each register bank contains 8 registers of 1 byte, in total 32 registers in register bank. By default register bank 0 is selected. RS1 and RS0 bits of PSW are used to select the register bank.

ii. 16 Byte-bit addressable area

In RAM there are 16 byte bit addressable areas, in which 128 bits can be accessed by suitable instruction. The address range of bit addressable area is 00h (LSB of first byte of 20h) to 7Fh (MSB of last byte of 2Fh)

iii. 80 byte-Scratch pad area & SFR.

The remaining 80 bytes are general purpose area which are shared with DATA and SFR (special function register) upper 128 byte of RAM accessed indirect addressing while SFR accessed by direct addressing only. SFR contain special register like ACC, B, IE, IP, IE etc. Below diagram shows architecture of RAM



7. Explain the Data types and Directives of 8051 Microcontroller

Data types: The 8051 microcontroller has only one data type. It is 8 bits, and the size of each register is also 8 bits. It is the job of the programmer to break down data larger than 8 bits (00 to FFH, or 0 to 255 in decimal) to be processed by the CPU. The data types used by the 8051 can be positive or negative

Directives

DB (define byte): The DB directive is the most widely used data directive in the assembler. It is used to define the 8-bit data. When DB is used to define data, the numbers can be in decimal, binary, hex, or ASCII formats. For decimal, the “D” after the decimal number is optional, but using “B” (binary) and “H” (hexadecimal) for the others is required.

Eg:-

	ORG 500H
Data1:	DB 28
Data2:	DB 01011100B
Data3:	DB 39H
Data4:	DB “microprocessors”

Assembler directives:

ORG (Origin): The ORG directive is used to indicate the beginning of the address. The number that comes after ORG can be either in hex or in decimal. If the number is not followed by H, it is decimal and the assembler will convert it to hex. Some assemblers use “. ORG” (notice the dot) instead of “ORG” for the origin directive

EQU (equate): This is used to define a constant without occupying a memory location. The EQU directive does not set aside storage for a data item but associates a constant value with a data label so that when the label appears in the program; its constant value will be substituted for the label

```
COUNT      EQU      25
.....
MOV R3, COUNT
```

END directive: Another important pseudo code is the END directive. This indicates to the assembler the end of the source (asm) file. The END directive is the last line of an 8051 program, meaning that in the source code anything after the END directive is ignored by the assembler

8. Write a program to create a square wave of 50% duty cycle on the P1.5 bit. Timer 0 is used to generate the time delay

```
#include <reg51.h>
void T0M1_delay (void);
```

```

sbit mybit=P1^5;
void main (void)
{ while(1)
  { mybit = ~mybit;
    T0M1_delay ();
  }}
void T0M1_delay (void)
{ TMOD=0x01;           // timer0 in mode 1(16-bit)
  TL0=0xFD;             // loading TLO and TH0, delay for 50ms
  TH0=0x4B;
  TR0=1;                // start the timer
  while (TF0==0);       // wait for TF0 to roll over
  TR0=0;    TF0=0;
}

```

PART B

Answer any two full questions, each carries 10 marks

9 a) Write an ALP in 8085 to find the smallest number in an array

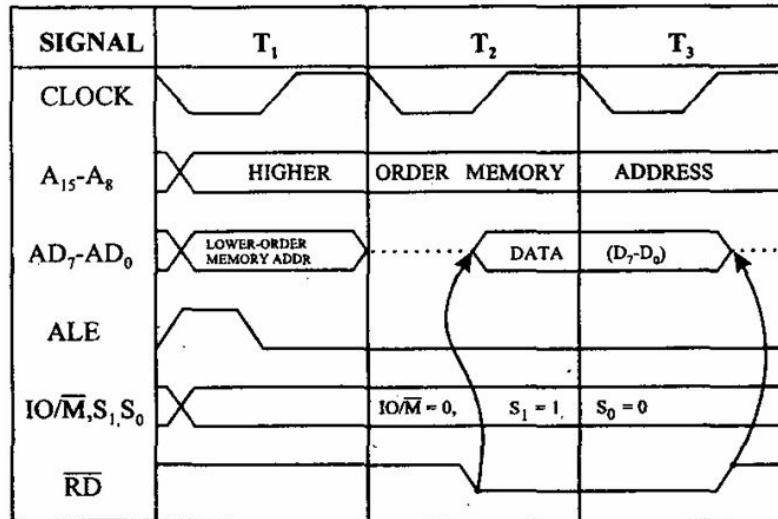
Program

```

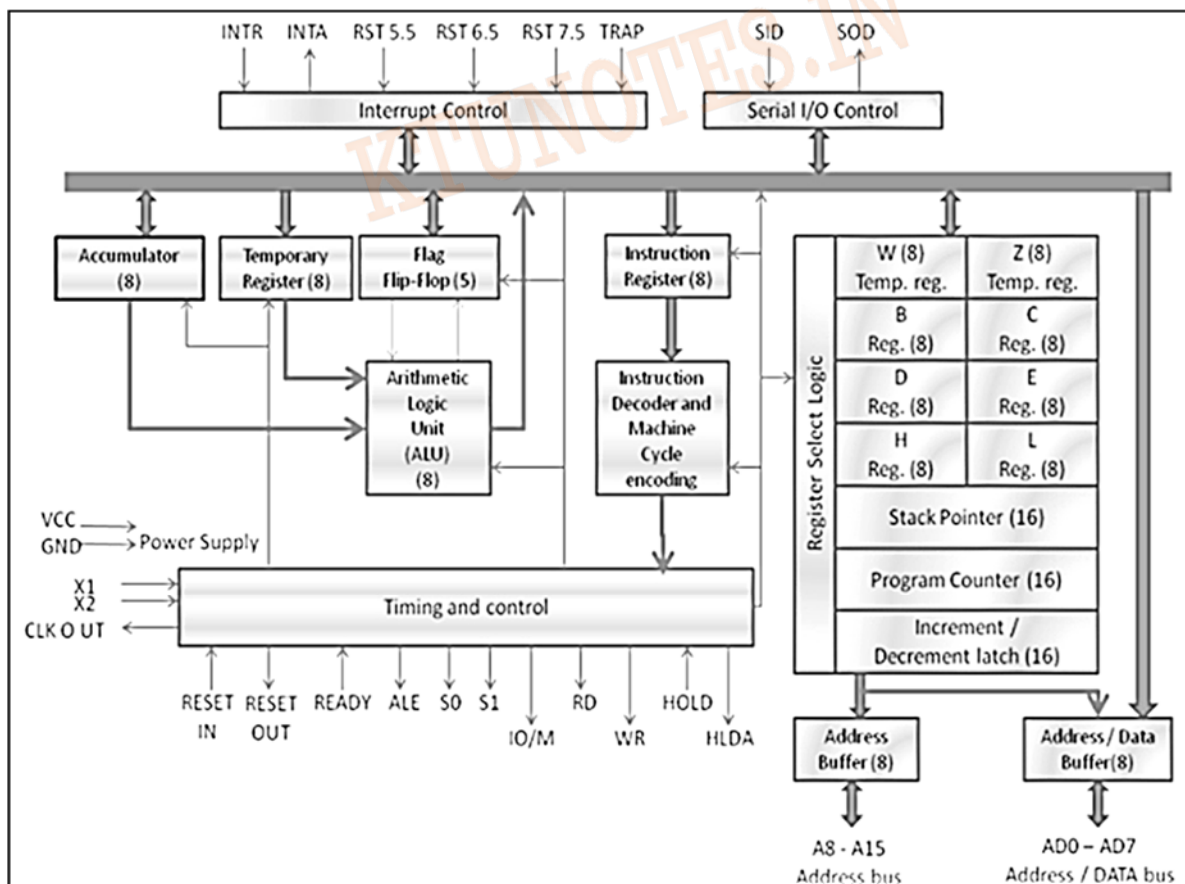
      LXI H, 5000H    ;Set pointer for array
      MOV B,M        ;Load the Count
      INX H
      MOV A,M        ;Set 1st element as largest data
      DCR B          ;Decrement the count
LOOP:  INX H
      CMP M          ;If A- reg < M go to AHEAD
      JC AHEAD
      MOV A,M        ;Set the new value as smallest
AHEAD: DCR B
      JNZ LOOP       ;Repeat comparisons till count = 0
      STA 6000        ;Store the largest value at 6000
      HLT

```

9 b) Draw the timing diagram for Memory Read operation



10 Explain with neat block diagram the architecture of Intel 8085 Microprocessor



Control Unit

Generates signals within uP to carry out the instruction, which has been decoded. In reality causes certain connections between blocks of the uP to be opened or closed, so that data goes where it is required, and so that ALU operations occur.

Arithmetic Logic Unit

The ALU performs the actual numerical and logic operation such as 'add', 'subtract', 'AND', 'OR', etc. Use data from memory and from Accumulator to perform arithmetic. Always stores result of operation in Accumulator.

Registers

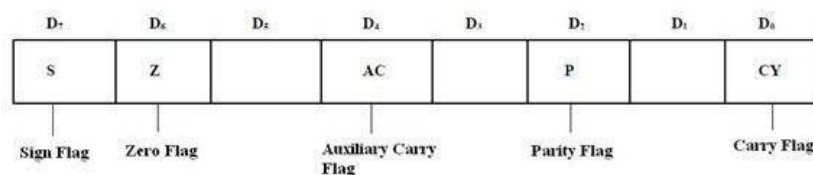
The 8085/8080A-programming model includes six registers, one accumulator, and one flag register, as shown in Figure. In addition, it has two 16-bit registers: the stack pointer and the program counter. They are described briefly as follows. The 8085/8080A has six general-purpose registers to store 8-bit data; these are identified as B,C,D,E,H, and L as shown in the figure. They can be combined as register pairs - BC, DE, and HL - to perform some 16-bit operations. The programmer can use these registers to store or copy data into the registers by using data copy instructions.

Accumulator

The accumulator is an 8-bit register that is a part of arithmetic/logic unit (ALU). This register is used to store 8-bit data and to perform arithmetic and logical operations. The result of an operation is stored in the accumulator. The accumulator is also identified as register A.

Flags

The ALU includes five flip-flops, which are set or reset after an operation according to data conditions of the result in the accumulator and other registers. They are called Zero(Z), Carry (CY), Sign (S), Parity (P), and Auxiliary Carry (AC) flags; their bit positions in the flag register are shown in the Figure below. The most commonly used flags are Zero, Carry, and Sign. The microprocessor uses these flags to test data conditions.



For example, after an addition of two numbers, if the sum in the accumulator is larger than eight bits, the flip-flop used to indicate a carry -- called the Carry flag (CY) -- is set to one. When an arithmetic operation results in zero, the flip-flop called the Zero (Z) flag is set to one. The first Figure shows an 8-bit register, called the flag register, adjacent to the accumulator. However, it is not used as a register; five bit positions out of eight are used to store the outputs of the five flip-flops. The flags are stored in the 8-bit register so that the programmer can examine these flags (data conditions) by accessing the register through an instruction.

These flags have critical importance in the decision-making process of the microprocessor. The conditions (set or reset) of the flags are tested through the software instructions. For example, the instruction JC (Jump on Carry) is implemented to change the sequence of a program when CY flag is set. The thorough understanding of flag is essential in writing assembly language programs.

Program Counter (PC)

This 16-bit register deals with sequencing the execution of instructions. This register is a memory pointer. Memory locations have 16-bit addresses, and that is why this is a 16-bit register.

The microprocessor uses this register to sequence the execution of the instructions. The function of the program counter is to point to the memory address from which the next byte is to be fetched. When a byte (machine code) is being fetched, the program counter is incremented by one to point to the next memory location.

Stack Pointer (SP)

The stack pointer is also a 16-bit register used as a memory pointer. It points to a memory location in R/W memory, called the stack. The beginning of the stack is defined by loading 16-bit address in the stack pointer. The stack concept is explained in the chapter "Stack and Subroutines."

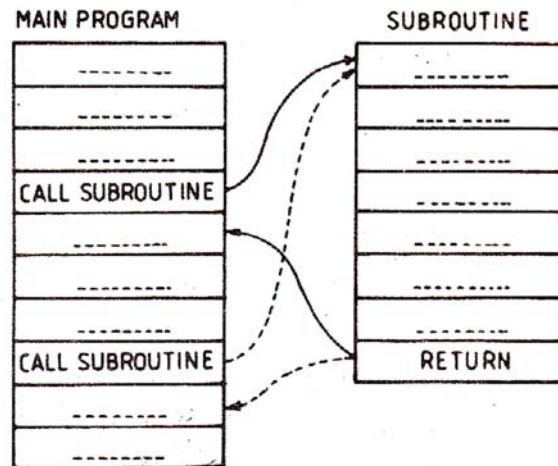
Instruction Register/Decoder

Temporary store for the current instruction of a program. Latest instruction sent here from memory prior to execution. Decoder then takes instruction and 'decodes' or interprets the instruction. Decoded instruction then passed to next stage.

11 a) Explain CALL & RETURN instructions in 8085

CALL → used in main program to call subroutines

RET → used in the end of subroutine to return to the main program



CALL RETURN structure of subroutine.

Data transfer → During CALL instruction

Memory Address		
2000	LXI SP,2400H	;Initialize the stack pointer at 2400H
↓	↓	
2040	CALL 2070H	;Call the subroutine located at 2070H. This is
2041		; a 3-byte instruction
2042		
2043	NEXT INSTRUCTION	;The address of the next instruction following
↓	↓	; the CALL instruction
205F	HLT	;End of the main program
2070	First Subroutine	;Beginning of the subroutine
↓	↓	
	Instruction	
↓	↓	
207F	RET	;End of the subroutine

11. b) Explain Machine Cycle and Instruction Cycle

CPU fetches one Instruction from memory at a time & executes it. Execution of all the instruction one by one to produce final result

Instruction cycle à Necessary steps a CPU carried out

- 1) Fetching an instruction
- 2) Getting necessary data from memory
- 3) Execution --- constitute IC ---

Total time for IC = FC + EC

Fetch cycle is the steps carried out to fetch an opcode from memory, Time required for FC à Fixed slot time. Execution cycle steps carried out to get a data (if any) from memory for a specific operation in an instruction & Execution. Time required for EC à Variable à depends on the type of instruction

MACHINE CYCLE

Necessary steps carried out to perform the operation of accessing either memory or I/O device, constitute a machine cycle

In machine cycle à Processor performs one of the basic operations such as

- Opcode fetch cycle
- Memory read cycle
- Memory write cycle
- I/O read cycle
- I/O write cycle

In all these operation microprocessor accesses the memory or I/O device once. Instruction cycle consists of several Machine cycle

PART C

Answer any two full questions, each carries 10 marks.

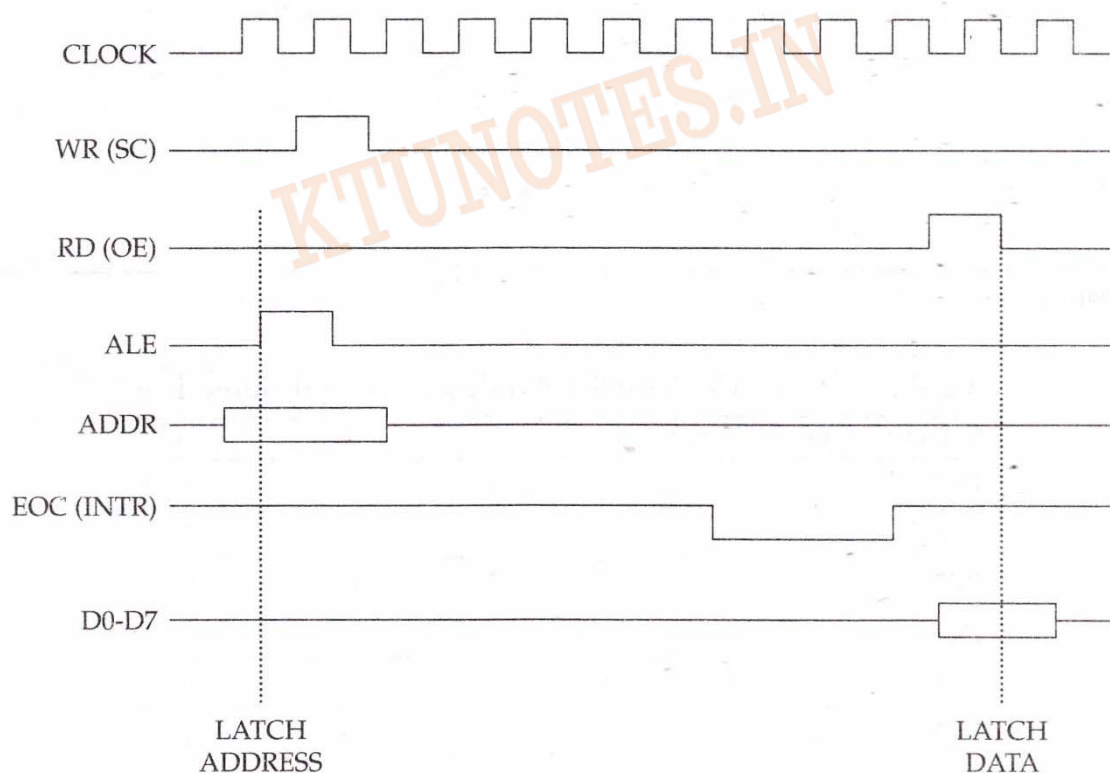
12 a) Explain with suitable diagram, how an ADC can be interfaced with 8085 microprocessor

Steps to program the ADC 0808

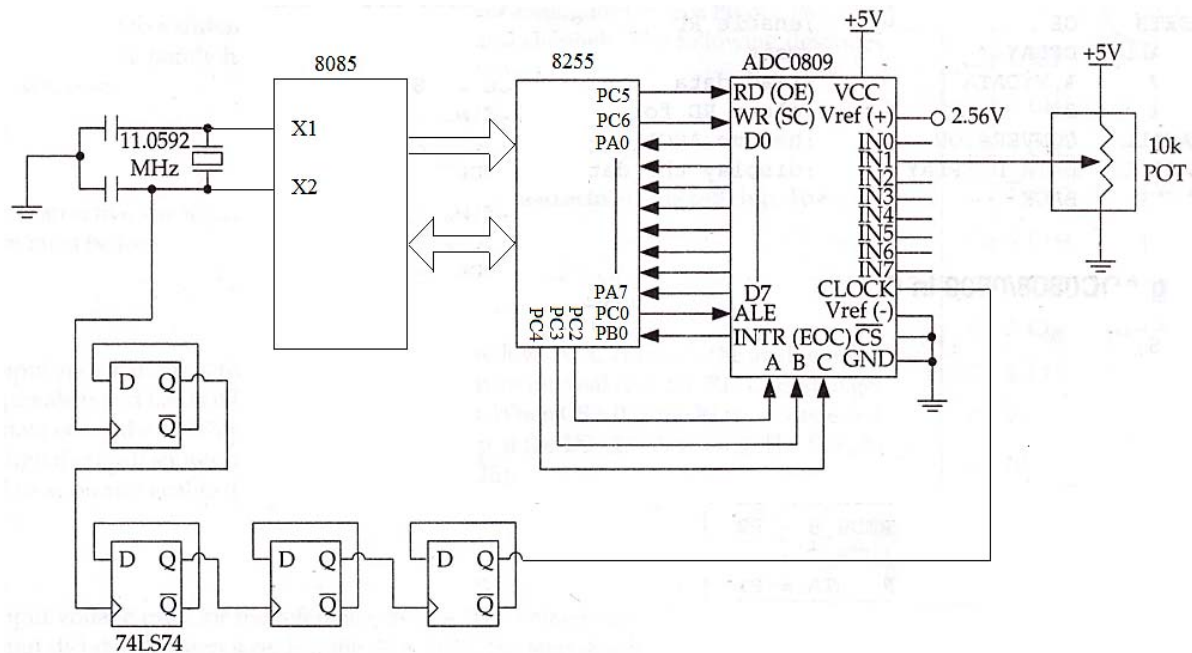
1. Select the analog channel by providing bits to A, B, and C
2. Activate the ALE pin. It needs a L-to- H pulse to latch the address
3. Activate SC (Start Conversion) by L-to-H pulse to initiate conversion
4. Monitor EOC (end of conversion) to see whether conversion is finished. H-to-L output indicates that the data is converted and ready to pick up.
5. Activate OE (output enable) to read data out of the ADC chip. A L-to-H pulse to the OE pin will bring digital data out of the chip

Note: - In ADC 0808, there is no self-clocking and clock must be provided from an external source. Also the conversion frequency is less compared to 8051 clock frequency

Timing Diagram



Interfacing of ADC 0808



Program

```

MVI A, 91H          ; make port A input, PB0 input pin
OUT 0FH
MVI A, 00H          ; clear ALE – PC0
OUT 0FH
MVI A, 0AH          ; clear OE – PC5
OUT 0FH
MVI A, 0CH          ; clear SC – PC6
OUT 0FH
back: MVI A, 05H      ; Select channel 1, PC2 = 1 (A)
OUT 0FH
MVI A, 06H          ; Select channel 1, PC3 = 0 (B)
OUT 0FH
MVI A, 08H          ; Select channel 1, PC4 = 0 (C)
OUT 0FH
CALL Delay
MVI A, 01H          ; SET ALE – PC0, and latch the address
OUT 0FH
CALL Delay
MVI A, 0DH          ; SET SC – PC6, start conversion
OUT 0FH
    
```

```

CALL Delay
MVI A, 00H          ; clear ALE – PC0
OUT 0FH
MVI A, 0CH          ; clear SC – PC6
OUT 0FH
again: IN 0DH        ; read port B to check PB0 is 1
ANI 01H
CPI 01H
JNZ again
repeat: IN 0DH       ; read port B to check PB0 is 0 ( H to L)
ANI 00H
CPI 00H
JNZ repeat
MVI A, 0BH          ; Set OE – PC5
OUT 0FH
CALL Delay
IN 0CH
Call Display
MVI A, 0AH          ; clear OE – PC5 for next data
OUT 0FH
JMP back

```

12. b) Explain Software and Hardware interrupts in 8085 Microprocessor

To answer a request from an I/O device, it is sometimes necessary to interrupt the execution of the main program. The CPU temporarily stops the ongoing process, attends the I/O device which has interrupted the CPU, then returns to what it was doing. Interrupt process of 8085 is controlled by the Interrupt Enable Flip Flop, which is internal to the microprocessor. Interrupt Enable Flip Flop can be set or reset by using software instructions

8051 interrupts are INTR, RST 7.5, RST 6.5, RST 5.5, TRAP

Interrupt Classification

- According upon their Maskability
 - iii. Maskable Interrupts → INTR, RST 7.5, RST 6.5, RST 5.5
 - iv. Non- maskable interrupts → TRAP
- According their address distribution method.

iii. Vectored Interrupts → Microprocessor automatically goes to specific address in response to an interrupt request without any external hardware

eg:- TRAP, RST 7.5, RST 6.5, RST 5.5

iv. Non-vectored Interrupts → Interrupt device has to give the address of the interrupt service routine

eg:- INTR

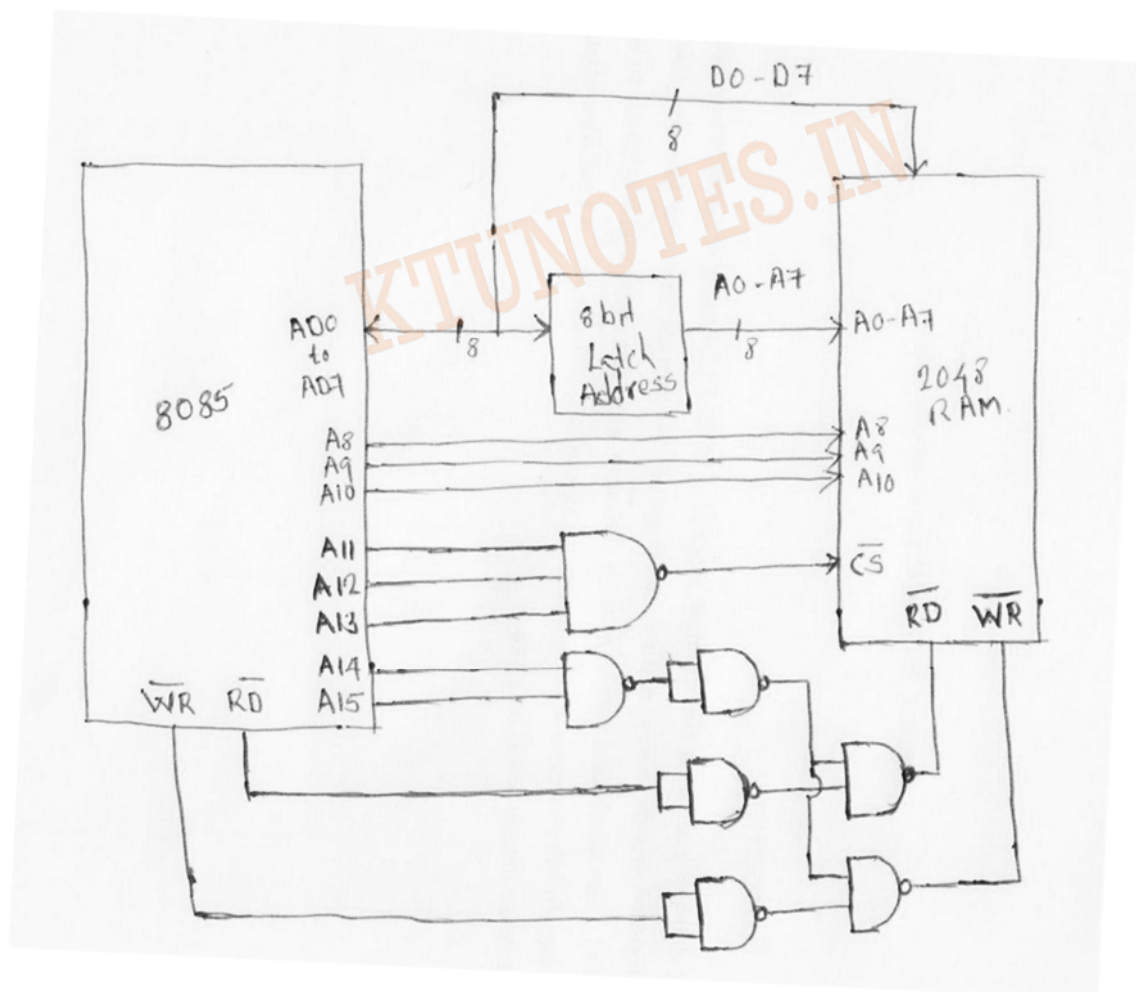
Again the interrupts can be classified into software and hardware interrupt

Software interrupt → INTR (using RST instructions)

Hardware interrupts → TRAP, RST 7.5, RST 6.5, RST 5.5

13 a) Draw an Interface circuit of 2048 x 8 bit RAM with 8085 using NAND Gate

Interfacing Diagram



Addressing Mapping

	A 15	A 14	A 13	A 12	A 11	A 10	A 9	A 8	A 7	A 6	A 5	A 4	A 3	A 2	A 1	A 0	Address in HEX
Starting Address	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	F800 H
Ending Address	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	FFFF H

13. b) Explain general characteristics of embedded system

- It should have **one or a small set of functions which it is expected to perform efficiently**
- It should be designed for **low-power dissipation**, because many systems are **battery powered**.
- It has **limited memory** and **limited number of peripherals**
- **Applications are not meant to be alterable by the user**
- Many of them are **not accessible directly**, that is, they **may be part of the control unit of a larger system**, so no interference in operation is possible.
- They need to be **highly reliable**
- Many of them **need to operate with time constraints**.

14 a) Explain i) Compiler ii) Assembler iii) Linker iv) Loaders

i) Compiler

- A compiler is a **program that translates one computer language program** (called source code) **into another** (object code).
- Usually the name 'compiler' is use for **programs that translate source code from a high-level language to a lower level language** (assembly language or machine code.)
- **For a machine to 'run' the object code**, the final conversion should be into the **machine's machine code**.
- In that case, the conversion from a high level language to assembly code can be considered to be an inter-mediate step.

ii) Assembler

- This is the software **which converts an assembly program into a machine code on a one-to-one basis.**
- The term 'one to one' means that for a **specific mnemonic**, there is one and only one machine code.

iii) Linker

- This software does the **linking of many modules.** There may be a number of **source modules.**
- There may also be a **number of library functions** being used.
- For example, if we started with a C program, we might be using the **standard I/O or math library, which needs to be 'connected' to the object files.**
- By linking, that is, **combining all these, we get the executable file**, which is usually designated as the 'exe file'.
- It is this **file which will be 'loaded' to the processor memory to get executed.**

iv) Loaders

- In computing, a loader is the **part of all system that is responsible for loading programs and libraries.**
- It is one of the essential stages in the process of starting a program, **as it places programs into memory and prepares them for execution.**
- Loading a program **involves reading the contents of the executable file containing the program instructions into memory, and then carrying out other required preparatory tasks to prepare the executable for running.**
- Once loading is complete, the **system starts the program by passing control to the loaded program code.**

14. b) What are the demerits of Waterfall Model?

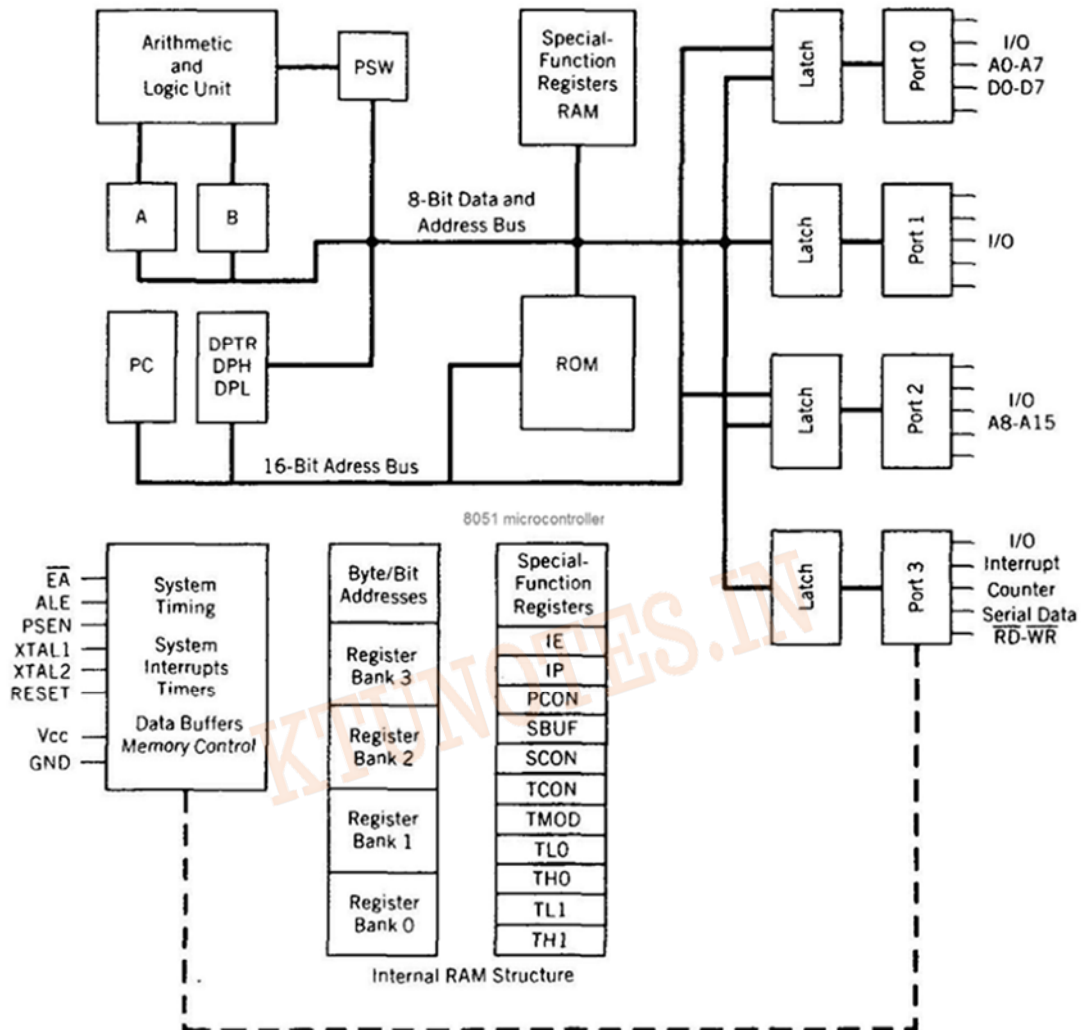
- **All requirements must be known before hand, as there is no phase which goes back and makes a correction.**
- **There is no opportunity for a customer to preview and correct the system. He gets the product at the end of all the phases, and this may be too late to incorporate any changes.**

PART D

Answer any two full questions, each carries 10 marks.

15. Explain with neat block diagram the architecture of 8051 Microcontroller.

8051 Architecture

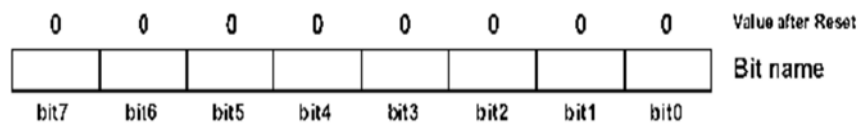


The architecture of 8051 contains

1. 8 bit CPU with registers A & B
2. Memory [Data/Program]
3. Input/output Ports
4. Special Function registers
5. Control Logic

A (Accumulator) Register:

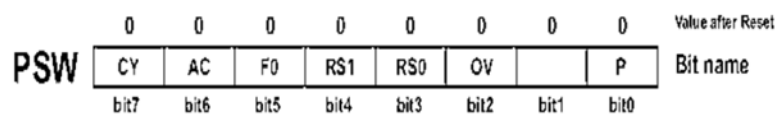
General purposes register which serves for storing data and immediate results for storing immediate results during operating.



B-Register

B-register is used during multiply and divide operation which can be performed only upon numbers stored in the A and B registers.

Program Status Word



PSW contains several status bits that reflect the current state of the CPU. This register contains: Carry bit, auxiliary carry, two register bank select bits, over flow flag, parity flag and user defined status flag.

P - Parity bit. If a number in accumulator is even then this bit will be automatically set (1), otherwise it will be cleared (0). It is mainly used during data transmission and receiving via serial communication. 1-ODD Parity, 0- EVEN Parity.

Bit 1: This bit is intended for the future versions of the microcontrollers, so it is not supposed to be here.

OV Overflow occurs when the result of arithmetical operation is greater than 255 (decimal), so that it cannot be stored in one register. In that case, this bit will be set (1). If there is no overflow, this bit will be cleared (0). it is also set to one in the case of Divided by Zero Occurs.

RS0, RS1 - Register bank selects bits. These two bits are used to select one of the four register banks in RAM. By writing zeroes and ones to these bits, a group of registers R0-R7 is stored in one of four banks in RAM

RS1	RS2	Space in RAM
0	0	Bank0 00h-07h
0	1	Bank1 08h-0Fh
1	0	Bank2 10h-17h
1	1	Bank3 18h-1Fh

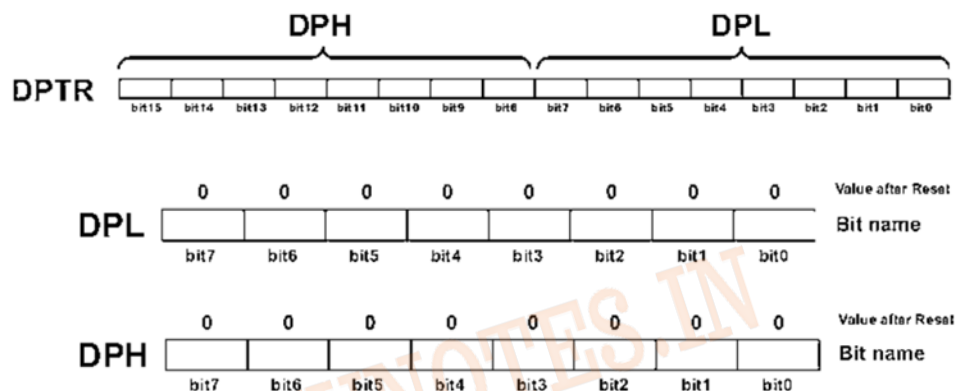
F0 - Flag 0. This is a general-purpose bit available to the user.

CY - Carry Flag is the (ninth) auxiliary bit used for all arithmetical operations and shift instructions.

AC - Auxiliary Carry Flag is used for BCD operations only. The auxiliary carry flag is set or cleared after an add instruction (*ADD A, operand* or *ADDC A, operand*) only. There are two conditions that result in AC being set. If the lower four bits of the accumulator (the lower nibble) are in the range AH to FH. If a carry was generated out of bit 3 into bit 4 of the accumulator.

DPTR Register

DPTR consist of two separate registers: DPH and DPL. It is used for external memory addressing. They may handle as a 16-bit register or as two 8-bit registers.



SP Register (Stack Pointer)

Pointer register for stack memory space of RAM, it stores the address of the stack top. SP is modified directly by the 8051 by six instructions: PUSH, POP, ACALL, LCALL, RET, and RETI. It is also used intrinsically whenever an interrupt is triggered.

The Program Counter (PC)

The Program Counter (PC) is a 16-bit register which stores the address of the next instruction to be executed. 8051 initialize always PC at 0000h and is incremented each time an instruction is executed. It is important to note that PC will not increment always by one. In some cases instructions require 2 or 3 bytes then PC will increment by 2 or 3.

Memory –

Internal ROM or EPROM (8751) of 0 to 4K

Internal RAM of 128 Bytes:

Four register banks, each containing 8 registers.

16 bytes, which may be addressed at the bit level.

80 bytes of general purpose data memory.

32 I/O pins arranged as four 8 bit ports: PO - P3

Two 16 bit timers / Counters - Timer0 & Timer1.

Full duplex serial data receiver / Transmit buffer: SBUF

Control registers: TCON, TMOD, SCON, PCON, IP & IE Five interrupts

Special Function Register (SFR) Memory

Special Function Registers (SFRs) are areas of memory that control specific functionality of the 8051 processor. That is, four SFRs permit access to the 8051's 32 input/output lines, some SFR allows a program to read or write to the 8051's serial port. Other SFRs allow the user to set the serial baud rate, control and access timers, and configure the 8051's interrupt system.

Eg:- TCON, TMOD, SCON, PCON, IP, IE, etc.

Features

- 4kB on chip program memory
- 128bytes on-chip data memory
- Four register bank
- 128 user defined software flags
- 64 Kbytes each program and external RAM addressability
- One microsecond instruction cycle with 12MHz crystal
- 2 multimode, 16 –bit Timers/ Counters
- Direct byte and bit addressability
- Binary or decimal arithmetic
- CPU with Boolean processor
- 5 or 6 interrupts: 2 are external
- 2 priority levels
- 2 or 3 16-bit timer/counters
- Programmable full-duplex serial port
- 32 I/O lines (four 8-bit ports)

16. a) Explain SFR registers in 8051 microcontroller

The Special function registers (SFR) in 8051 micro controller and this includes Register A, Register B, Processor Status Word (PSW), PCON etc etc. There are 21 unique locations for these 21 special function registers and each of these register is of 1 byte size. Some of these special function registers are bit addressable (which means you can access 8 individual bits inside a single byte), while some others are only byte addressable.

Register A/Accumulator

The most important of all special function registers. The Accumulator (sometimes referred to as Register A also) holds the result of most of arithmetic and logic operations. ACC is usually accessed by direct addressing and its physical address is E0H. Accumulator is both byte and bit addressable. To access the 1st bit (i.e bit 0) or to access accumulator as a single byte (all 8 bits at once), you may use the same physical address E0H. Also to access the second bit (ie bit 1), you may use E1H and for third bit E2H and so on.

Register B

The major purpose of this register is in executing multiplication and division. The 8051 micro controller has a single instruction for multiplication (MUL) and division (DIV).

Port Registers

There are 4 ports for 8051, named P0, P1, P2 and P3. Data must be written into port registers to send it out to any other external device through ports. Similarly any data received through ports must be read from port registers for performing any operation. All 4 port registers are bit as well as byte addressable.

The physical address of port 0 is 80

The physical address of port 1 is 90

And that of port 2 is A0

And that of port 3 is B0

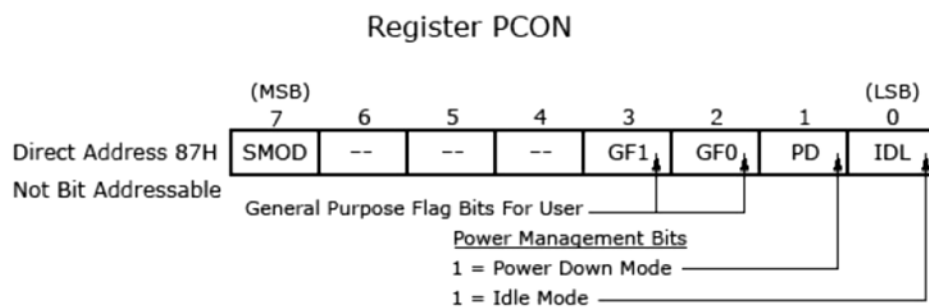
Stack Pointer

Known popularly with an acronym SP, stack pointer represents a pointer to the system stack. Stack pointer is an 8 bit register, the direct address of SP is 81H and it is only byte addressable. The content of the stack pointer points to the last stored location of system stack. To store something new in system stack, the SP must be incremented by 1 first and then execute the “store” command. Usually after a system reset SP is initialized as 07H and

data can be stored to stack from 08H onwards. This is usually a default case and programmer can alter values of SP to suit his needs.

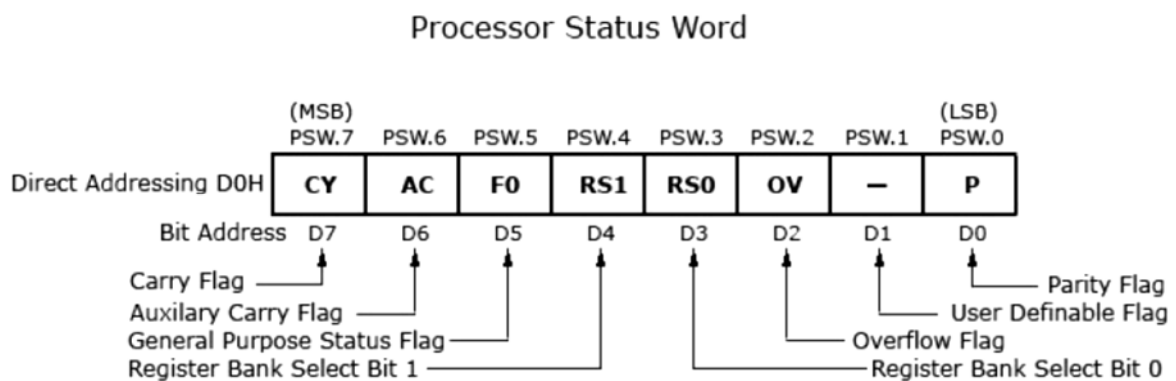
Power Management Register (PCON)

As the name indicates, this register is used for efficient power management of 8051 micro controller. Commonly referred to as PCON register, this is a dedicated SFR for power management alone. From the figure below you can observe that there are 2 modes for this register: - Idle mode and Power down mode.



Program Status Word

Commonly known as the PSW register, this is a vital SFR in the functioning of micro controller. This register reflects the status of the operation that is being carried out in the processor. The picture below shows PSW register and the way register banks are selected using PSW register bits – RS1 and RS0. PSW register is both bit and byte addressable. The physical address of PSW starts from D0H. The individual bits are explained below.



So far we have discussed about all major SFR's in 8051. The other SFR's are listed below table.

SFR	Address	Function
DPH	83	Data pointer registers (High). Only byte addressing possible.
DPL	82	Data pointer register (Low). Only byte addressing possible.
IP	B8	Interrupt priority. Both bit addressing and byte addressing possible.
IE	A8	Interrupt enable. Both bit addressing and byte addressing possible.
SBUF	99	Serial Input/Output buffer. Only byte addressing is possible.
SCON	98	Serial communication control. Both bit addressing and byte addressing possible.
TCON	88	Timer control. Both bit addressing and byte addressing possible.
TH0	8C	Timer 0 counter (High). Only byte addressing is possible.
TL0	8A	Timer 0 counter (Low). Only byte addressing is possible.
TH1	8D	Timer 1 counter (High). Only byte addressing is possible.
TL1	8B	Timer 1 counter (Low). Only byte addressing is possible.
TMOD	89	Timer mode select. Only byte addressing is possible.

16. b) Explain TMOD and TCON registers of 8051 Microcontroller

Timer Mode Control (TMOD) Register

The TMOD register is used to select the operating mode and the timer/counter operation of the timers.

The format of TMOD register is,

TMOD Address: 45C
Not Bit Addressable
Reset Value: 00H

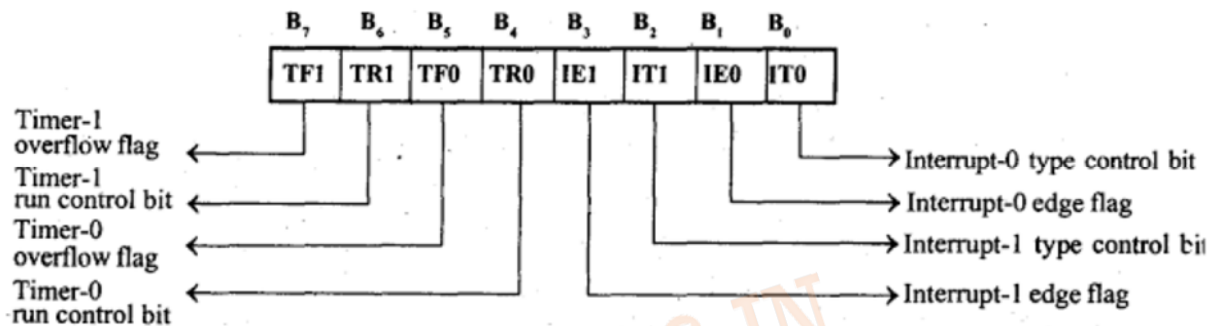


GATE	Gating control when set. Timer/Counter "n" is enabled only while $\overline{INTn}$ pin is high and "TRn" control bit is set. When cleared Timer "n" is enabled whenever "TRn" control bit is set.		
C/T	Timer or Counter Selector cleared for Timer operation (input from internal system clock.) Set for Counter operation (input from "Tn" input pin).		
M1	M0	OPERATING	
0	0	16-bit auto-reload timer/counter	
0	1	16-bit non-auto-reload timer/counter	
1	0	8-bit auto-reload timer/counter	
1	1	Dual 8-bit timer mode (timer 0 only)	

The lower four bits of TMOD register is used to control timer-0 and the upper four bits are used to control timer-1.

- The two timers can be independently program to operate in various modes.
- The TMOD register has two separate two bit field M0 and M1 to program the operating mode of timers. The operating modes of timers are mode-0, mode-1, mode-2 and mode-3. In all these operating modes the oscillator clock is divided by 12 and applied as input clock to time

TCON (Timer Control register)



The TCON register has a 1-bit flag, TF for each timer to indicate the timer overflow or end of timing. Whenever the timer/counter overflows, the TF flag is set to one. The TF flag is also used as an interrupt signal to initiate the execution of a subroutine. When the controller executes a subroutine, the TF flag is cleared. The TR bit is used to start/stop the timer/counter. When TR bit is set to one, the timer/ counter will start counting and continue the counting as long as TR bit is one. The timer/counter will stop counting when TR bit is cleared to zero. When a valid external interrupt signal is detected the IE flag is set to one. When the controller accepts the external interrupt and start processing it, the IE flag is cleared to zero. The IT = 1, when it recognize falling edge triggered external interrupt and IT = 0, when it recognize logic low level external interrupt.

17. Write an ALP in 8051 to add two 32 bit numbers & store the result

Let's assume that the numbers are stored at RAM location 40-43H and 50-53H respectively.

```
ORG 0H
MOV 40H, #23
MOV 41H, #15
MOV 42H, #60
MOV 43H, #70
```

```
MOV 50H, #30
MOV 51H, #40
MOV 52H, #63
MOV 53H, #77
```

```
MOV R0, #40H      ;pointer of bank 0
MOV R1, #60H      ;result storage bank 0
MOV R2, #4         ;counter
SETB RS0
MOV R1, #50H      ;pointer bank 1
CLR RS0
CLR C
```

```
REPEAT: MOV A, @R0
        SETB RS0
        ADDC A, @R1
        INC R1
        CLR RS0
        MOV @R1, A
        INC R0
        INC R1
        DJNZ R2, REPEAT
        JNC EXIT
        INC @R1
```

```
EXIT:   NOP
        SJMP $
```

```
END
```